

Goals:

- Definition of topological space;
- Examples of topological spaces.

Review:

- Open sets in metric spaces and their properties;
- Collections of sets;
- De Morgan's laws.

Definition. Let X be a nonempty set. A topology on X is a collection $\mathcal{T}$ of subsets of X such that

- i) $\emptyset, X \in \mathcal{T}$ (the empty set and the whole space are open)
- ii) If $A_1, A_2 \in \mathcal{T}$, then $A_1 \cap A_2 \in \mathcal{T}$ (the intersection of two open sets is open)
- iii) If $\{A_i\}_{i \in I} \subseteq \mathcal{T}$, then $\bigcup_{i \in I} A_i \in \mathcal{T}$. (the arbitrary union of open sets is open)

The pair $(X, \mathcal{T})$ is called **topological space** and the sets in the collection $\mathcal{T}$ are called **open sets**.

NOTE: (ii) is equivalent to (ii') if $A_1, \dots, A_n \in \mathcal{T}$, then $A_1 \cap \dots \cap A_n \in \mathcal{T}$.

Example 1 Let X be a nonempty set. The following are topologies on X

- a) The **indiscrete** or **trivial** topology $\mathcal{T} = \{\emptyset, X\}$

The only open subsets are $\emptyset$ and X

- b) The **discrete** topology $\mathcal{T} = \mathcal{P}(X)$

Every subset of X is open.

- c) The **cofinite** topology $\mathcal{T} = \{A \subseteq X \mid X \setminus A \text{ is finite}\} \cup \{\emptyset\}$

proof that $\mathcal{T}$ is a topology

• Note that $\emptyset \in \mathcal{T}$ and $X \setminus X = \emptyset$ is finite, so $X \in \mathcal{T}$.

• let $A_1, A_2 \in \mathcal{T}$. Then A_1^c and A_2^c are finite.

By De Morgan's law $(A_1 \cap A_2)^c = A_1^c \cup A_2^c$, which is finite.

Therefore $A_1 \cap A_2 \in \mathcal{T}$

• let $\{A_i\}_{i \in I} \subseteq \mathcal{T}$. Then, A_i^c is finite $\forall i \in I$

By De Morgan's law $(\bigcup_{i \in I} A_i)^c = \bigcap_{i \in I} A_i^c$, which is finite

Therefore $\bigcup_{i \in I} A_i \in \mathcal{T}$.

So $\mathcal{T}$ is a topology ~~✗~~

Example 2 Let $X \neq \emptyset$. If d is a metric on X , then the collection of all sets that are open according to the definition we gave in a metric space is a topology on X . More precisely the topology is

$$\mathcal{T} = \{A \subseteq X \mid \forall x \in A, \exists r > 0 \text{ s.t. } B_r(x) \subseteq A\}.$$

This is called the **topology induced by the metric d** .

Definition. We say that a topological space $(X, \mathcal{T})$ is **metrizable** if there exists a metric on X that induces the topology $\mathcal{T}$.

Note: if a topological space is metrizable, the topology is not induced by a unique metric. Indeed, in this case, there are always infinitely many distinct metrics that induce the same topology (this will be a homework exercise)

Exercise 1 Prove that a discrete topological space is metrizable.

proof Let $X \neq \emptyset$ and let d be the discrete metric on X .

Let $A \subseteq X$. Then A is open (hw exercise)

This shows that the topology induced by the discrete metric is the collection of all subsets of X , i.e. the discrete topology.

~~✗~~

Definition. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on a nonempty set X . If $\mathcal{T}_1 \subseteq \mathcal{T}_2$ we say that $\mathcal{T}_2$ is finer than $\mathcal{T}_1$ (or that $\mathcal{T}_1$ is coarser than $\mathcal{T}_2$).

finer = more open sets

coarser = less open sets

Exercise 2 Given $X \neq \emptyset$ is there a coarsest topology? What about a finest topology?

The indiscrete topology is the coarsest top. on a set X ,

The discrete topology is the finest top. on a set X .

Indeed, for any topology $\mathcal{T}$ on X we have

$$(\text{indiscrete topology}) = \{\emptyset, X\} \subseteq \mathcal{T} \subseteq \mathcal{P}(X) (= \text{discrete top})$$

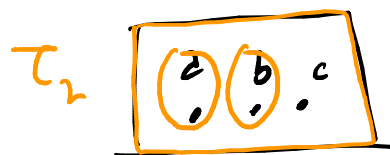
Exercise 3 Let $X = \{a, b, c\}$. Determine which of the following are topologies on X . Compare those that are two by two to determine which are finer or coarser.

a) $\mathcal{T}_1 = \{\emptyset, X, \{a\}\}$

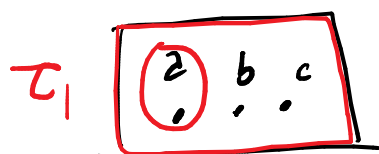
b) $\mathcal{T}_2 = \{\emptyset, X, \{a\}, \{b\}\}$

c) $\mathcal{T}_3 = \{\emptyset, X, \{a, b\}\}$

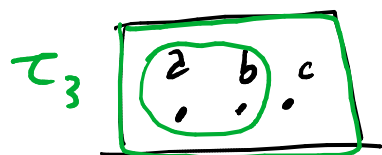
d) $\mathcal{T}_4 = \{\emptyset, X, \{a\}, \{a, b\}, \{a, b, c\}\}$



$\mathcal{T}_2$ is NOT a topology on X , because $\{a\}, \{b\} \in \mathcal{T}_2$ but $\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_2$.



$\mathcal{T}_1$ is a topology on X



$\mathcal{T}_3$ is a topology on X



$\mathcal{T}_4$ is a topology on X

$\mathcal{T}_4$ is finer than $\mathcal{T}_1$ and $\mathcal{T}_3$ since $\mathcal{T}_1 \subseteq \mathcal{T}_4$ and $\mathcal{T}_3 \subseteq \mathcal{T}_4$.

$\mathcal{T}_1$ and $\mathcal{T}_3$ are not comparable since $\mathcal{T}_1 \not\subseteq \mathcal{T}_3$ and $\mathcal{T}_3 \not\subseteq \mathcal{T}_1$.

Example 3 The collection $\mathcal{T} = \{]-\infty, x[\mid x \in \mathbb{R} \} \cup \{ \emptyset, \mathbb{R} \}$ is a topology on $\mathbb{R}$ that is (finer or coarser?) than the Euclidean topology.

You will show $\mathcal{T}$ is a topology as an exercise.

Denote by $\mathcal{E}$ the Euclidean topology on $\mathbb{R}$.

We know that any set of the form $] -\infty, x[$ for $x \in \mathbb{R}$

is open in the Euclidean topology, thus $\mathcal{T} \subseteq \mathcal{E}$.

This means $\mathcal{T}$ is coarser than $\mathcal{E}$. Moreover, sets of

the form $]a, b[$ with $a, b \in \mathbb{R}$, $a < b$ are open in the

Euclidean topology but are not in $\mathcal{T}$, so $\mathcal{T} \subsetneq \mathcal{E}$ ($\mathcal{T}$ is strictly coarser than $\mathcal{E}$).

Exercise 4 Determine if $\mathcal{T} = \{]a, b[\mid a, b \in \mathbb{R}, a < b \} \cup \{ \emptyset, \mathbb{R} \}$ is a topology on $\mathbb{R}$. If it is, determine whether it is comparable to the Euclidean one.

It is not. Indeed $]0, 1[\in \mathcal{T}$ and $]2, 3[\in \mathcal{T}$

but $]0, 1[\cup]2, 3[\notin \mathcal{T}$.

